

# DecisionIQ Master Handbook

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## CHAPTER 12: Machine Learning Pipelines (Churn, Revenue Forecasting, and Supplier Risk)

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### 12.1 Customer Churn Predictor

- **Target:** `is_churn_risk` (1 if Churned or At-Risk, 0 if Active).
  - **Algorithm:** Random Forest Classifier.
  - **Features:** Recency (days since last purchase), purchase frequency, monetary value, CSAT satisfaction rating, support ticket count, segment, and region.
  - **Feature Importance:** The model ranks features by predictive value (Gini importance):
    - *Recency & Average Order Value:* High importance (shows drop in engagement).
    - *CSAT & Tickets:* Low importance (indicates complaints are secondary to transaction frequency).
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### 12.2 Weekly Revenue Forecaster

- **Target:** Next 12 weeks of revenue.
  - **Algorithm:** Autoregressive Ridge Regression.
  - **Features:**
    - *Trend index:* Increments weekly to model general business growth.
    - *Lags:* Weekly revenue at  $t - 1$ ,  $t - 2$ ,  $t - 3$ ,  $t - 4$ .
    - *Rolling Mean:* 4-week moving average.
    - *Seasonality:* Week-of-year and month-of-year integers.
  - **Evaluation:** Evaluated using Mean Absolute Error (MAE) and R-squared.
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### 12.3 Supplier Logistics Delay Risk Classifier

- **Target:** Probability of shipment arriving late (`delivery_status == 'Late'`).
- **Algorithm:** Random Forest Classifier.
- **Features:** Supplier ID, warehouse ID, carrier, and product category.
- **Scorecard:** The model estimates late shipment probabilities, alerting operations managers which pending orders are at high risk.